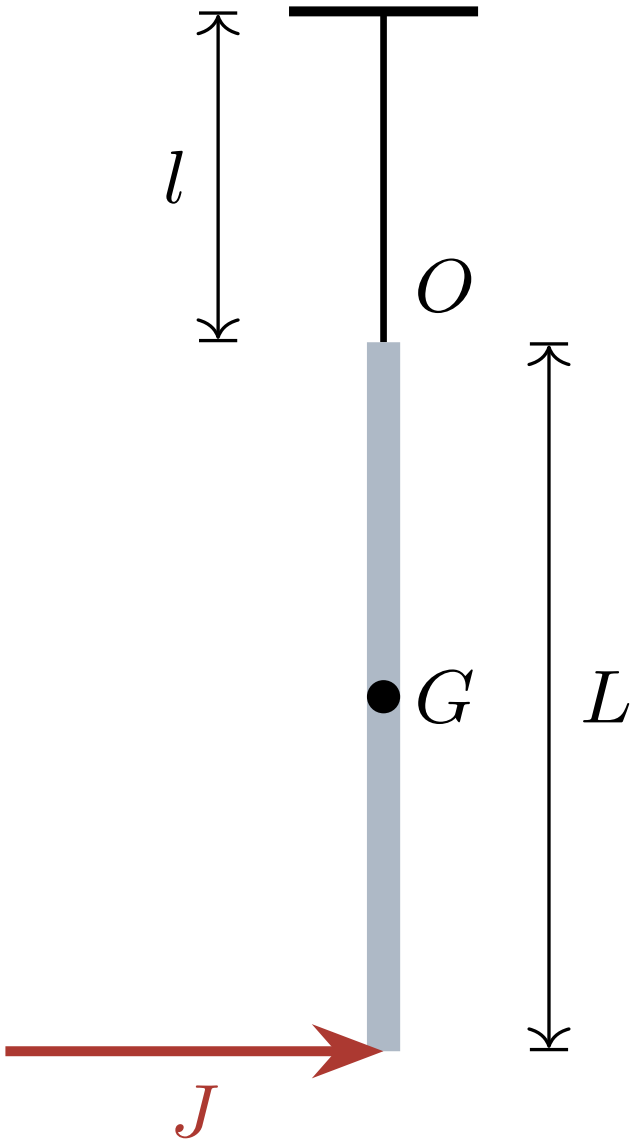
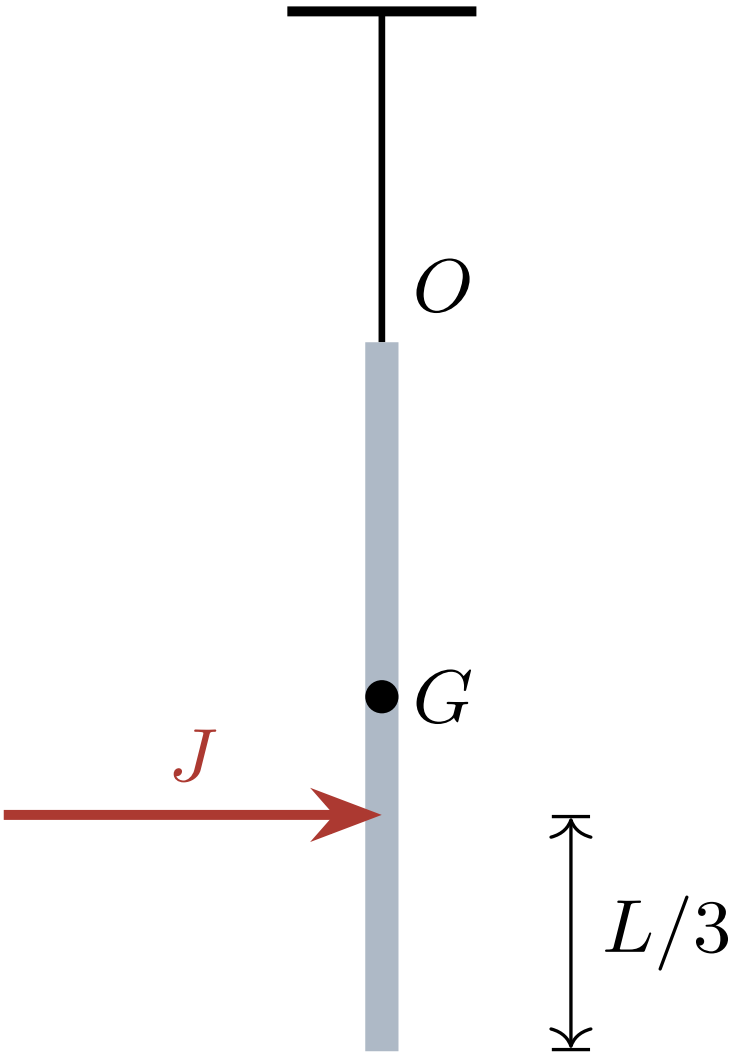
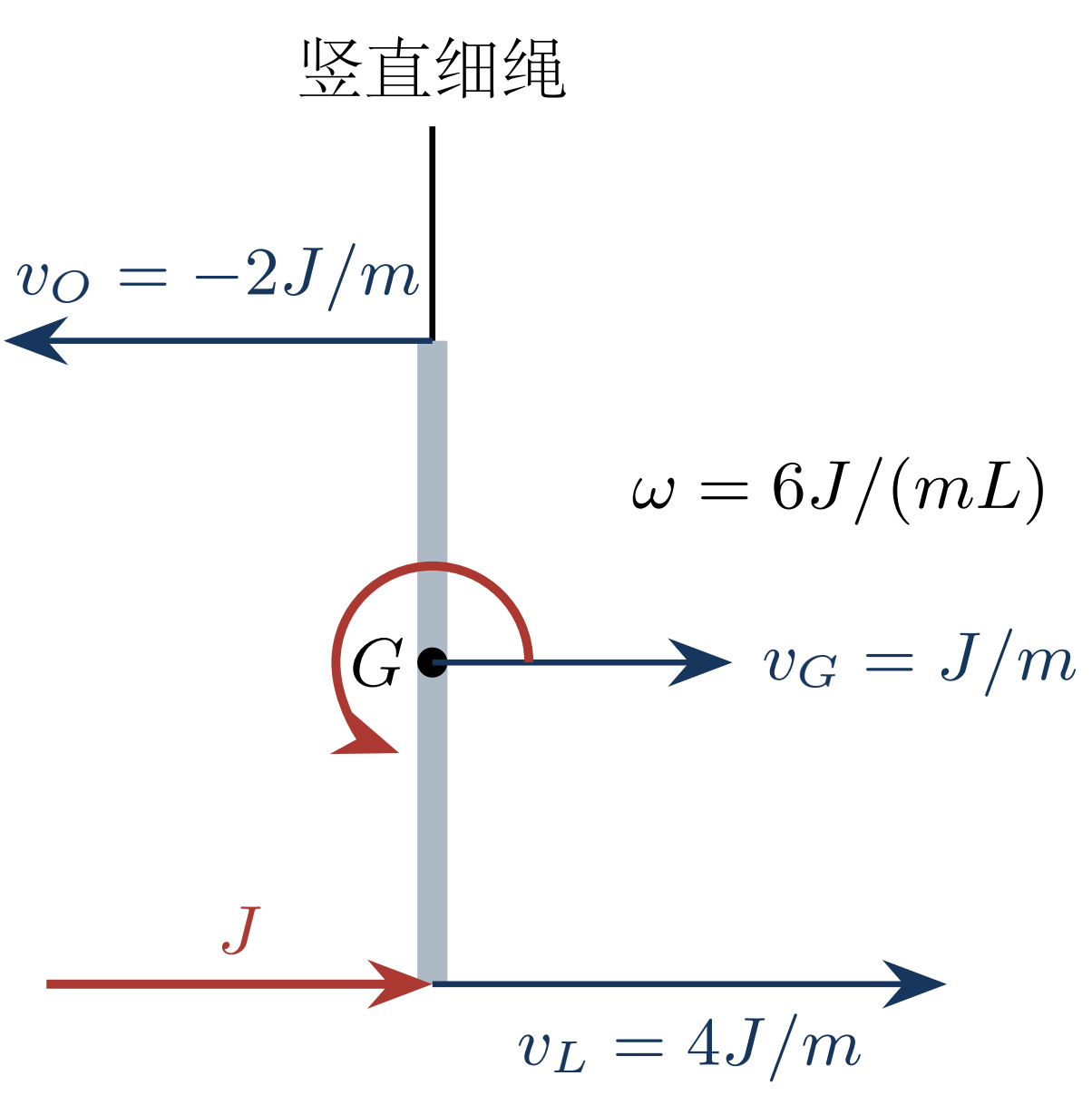
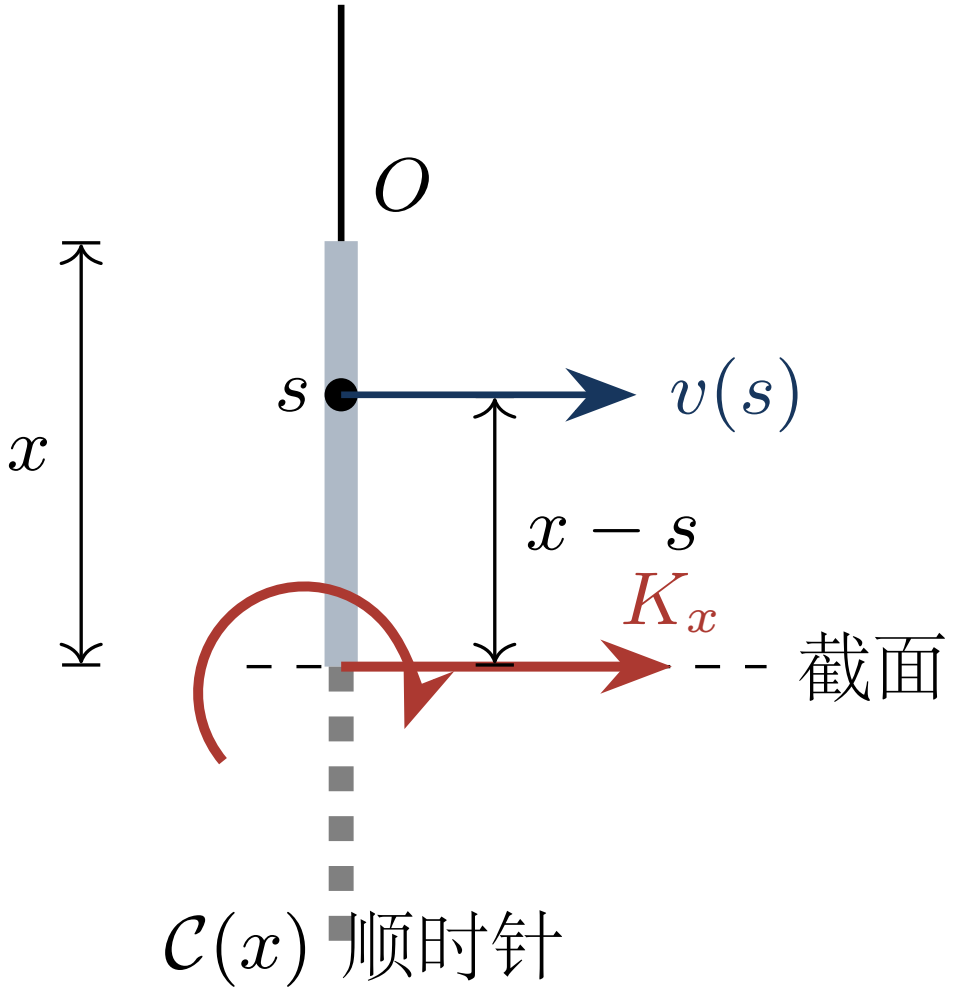


底端打击

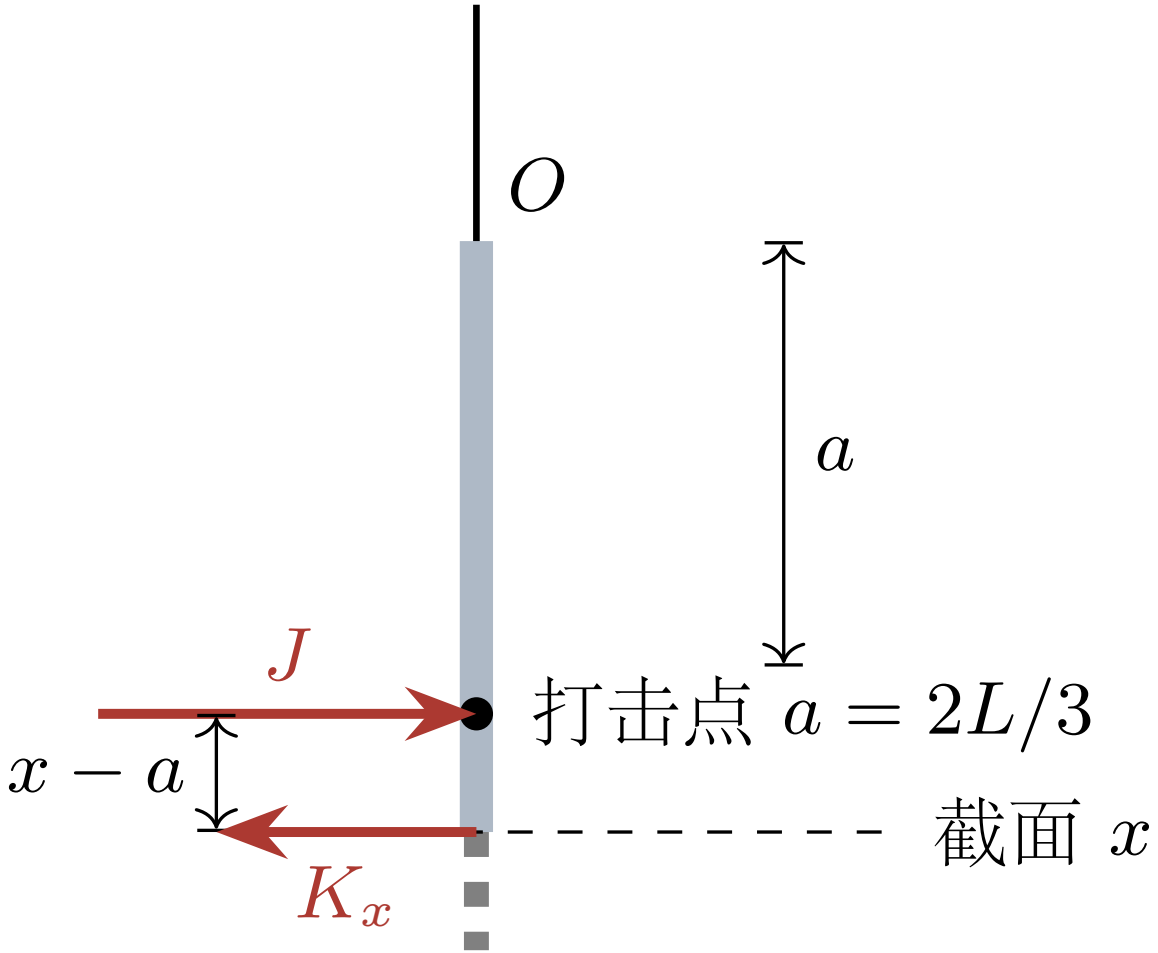


距底端 $L/3$ 打击

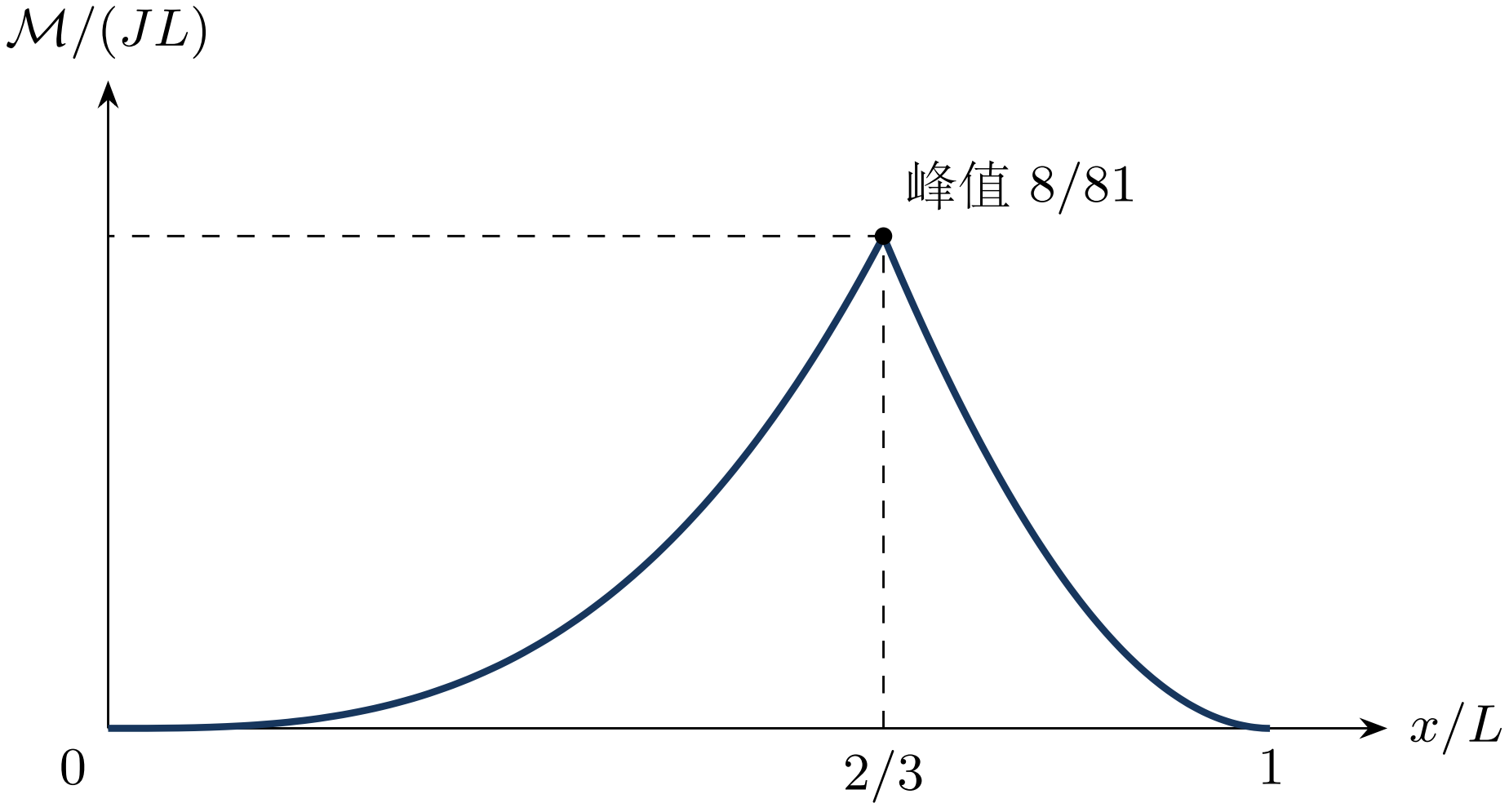

竖直细绳
$v_O = -2J/m$
$\omega = 6J/(mL)$
G
$v_G = J/m$
J
$v_L = 4J/m$



$x < a$，上段未受 J



上段包含外冲量 J


$\mathcal{M}/(JL)$
峰值 8/81
x/L
0
2/3
1